E1 244 Detection and Estimation Theory
Assignment 4

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Problem 1

Ans:

The state-space model is:

$$y_i = x_i + w_i, \quad i = 1, 2, \ldots, N, \quad w_i \sim \mathcal{N}(0, \sigma_w^2),$$

$$x_i = \alpha\, x_{i-1}, \qquad x_0 = \theta \sim \mathcal{N}(0, \sigma^2),$$

with θ independent of $w_1, w_2, \ldots, w_N$.

(a) MMSE estimate $\hat{\theta}_{\text{MMSE}}$

Since $x_i = \alpha^i \theta$, the signal at time i is $x_i = \alpha^i \theta$. Therefore:

$$p(y_i \mid \theta) \sim \mathcal{N}(\alpha^i \theta,\, \sigma_w^2).$$

The joint likelihood (observations are conditionally independent given θ) is:

$$p(\mathbf{y} \mid \theta) = \frac{1}{(2\pi)^{N/2}\, \sigma_w^N} \exp\left(-\frac{1}{2\sigma_w^2} \sum_{i=1}^{N} (y_i - \alpha^i \theta)^2 \right).$$

The prior is $p(\theta) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{\theta^2}{2\sigma^2}\right)$.

To compute $p(\mathbf{y}) = \int_{-\infty}^{\infty} p(\mathbf{y} \mid \theta)\, p(\theta)\, d\theta$, we evaluate the non-constant part of the integrand:

$$\int_{-\infty}^{\infty} \exp\left[-\frac{1}{2\sigma_w^2} \left(\sum_i y_i^2 + \theta^2 \sum_i \alpha^{2i} - 2\theta \sum_i \alpha^i y_i \right) \right] \exp\left(-\frac{\theta^2}{2\sigma^2} \right) d\theta.$$

Grouping terms and defining:

$$A = \frac{\sum_i \alpha^{2i}}{\sigma_w^2} + \frac{1}{\sigma^2}, \qquad B = \frac{\sum_i \alpha^i y_i}{\sigma_w^2}, \qquad C = \frac{\sum_i y_i^2}{\sigma_w^2},$$

the integrand becomes $\exp\left[-\frac{1}{2}(A\theta^2 - 2B\theta + C)\right]$. Completing the square:

$$\int_{-\infty}^{\infty} \exp\left[-\frac{A}{2}\left(\theta - \frac{B}{A}\right)^2\right] \exp\left[-\frac{1}{2}\left(C - \frac{B^2}{A}\right)\right] d\theta = \sqrt{\frac{2\pi}{A}} \exp\left[-\frac{1}{2}\left(C - \frac{B^2}{A}\right)\right].$$

Therefore:

$$E(\theta \mid \mathbf{y}) = \frac{B}{A} = \frac{\dfrac{\sum_i \alpha^i y_i}{\sigma_w^2}}{\dfrac{\sum_i \alpha^{2i}}{\sigma_w^2} + \dfrac{1}{\sigma^2}} = \frac{\sum_i \alpha^i y_i}{\sum_i \alpha^{2i} + \dfrac{\sigma_w^2}{\sigma^2}}.$$

$$\boxed{\hat{\theta}_{\text{MMSE}}(\mathbf{y}) = \frac{\sum_{i=1}^N \alpha^i y_i}{\sum_{i=1}^N \alpha^{2i} + \dfrac{\sigma_w^2}{\sigma^2}}.}$$

(b) Recursive LLSE

Since $\hat{\theta}_{\text{MMSE}}$ is linear in $\mathbf{y}$, it is also the LLSE. We derive it recursively.

Initialization ($i = 1$):

$$\hat{\theta}_1 = l_1\, y_1, \quad l_1 = \frac{E[\theta y_1]}{E[y_1^2]} = \frac{E[\theta(\alpha\theta + w_1)]}{E[(\alpha\theta + w_1)^2]} = \frac{\alpha\sigma^2}{\alpha^2\sigma^2 + \sigma_w^2}.$$

Recursive update:

$$\hat{\theta}_i = \hat{\theta}_{i-1} + l_i\, z_i, \quad z_i = y_i - \hat{y}_i(y_1, \ldots, y_{i-1}). \tag{1}$$

Since $y_i = \alpha^i\theta + w_i$ and from LLS properties $\hat{w}_i = 0$ (as θ, w_i, and w_j for $i \neq j$ are independent with zero means):

$$\hat{y}_i(y_1, \ldots, y_{i-1}) = \alpha^i \hat{\theta}_{i-1}, \qquad \Rightarrow \qquad z_i = y_i - \alpha^i \hat{\theta}_{i-1}.$$

The gain $l_i = \dfrac{E[\theta\, z_i]}{E[z_i^2]}$. Denoting the error $e_i = \theta - \hat{\theta}_i$:

$$E[\theta\, z_i] = E\Big[\theta(y_i - \alpha^i\hat{\theta}_{i-1})\Big] = E\Big[\alpha^i\theta(\theta - \hat{\theta}_{i-1}) - \theta w_i\Big] = \alpha^i E[e_{i-1}^2]$$

(using $E[\theta w_i] = 0$ and orthogonality of error and estimate).

$$E[z_i^2] = E\big[(\alpha^i e_{i-1} + w_i)^2\big] = \alpha^{2i} E[e_{i-1}^2] + \sigma_w^2$$

(using $E[e_{i-1} w_i] = 0$, since e_{i-1} depends on $y_1, \ldots, y_{i-1}$ which are independent of w_i).

Therefore:

$$l_i = \frac{\alpha^i E[e_{i-1}^2]}{\alpha^{2i} E[e_{i-1}^2] + \sigma_w^2}.$$

The mean-squared error evolves as:

$$\begin{aligned} E[e_i^2] &= E\big[(e_{i-1} - l_i(\alpha^i e_{i-1} + w_i))^2\big] = (1 - l_i\alpha^i)^2 E[e_{i-1}^2] + l_i^2\,\sigma_w^2 \\ &= \frac{\sigma_w^2\, E[e_{i-1}^2]}{\alpha^{2i} E[e_{i-1}^2] + \sigma_w^2}. \end{aligned}$$

Define $K_i = \dfrac{\sigma_w^2}{E[e_i^2]}$ with $K_0 = \dfrac{\sigma_w^2}{\sigma^2}$ (since $\hat{\theta}_0 = 0 \Rightarrow E[e_0^2] = \sigma^2$). Then:

$$K_i = \frac{\sigma_w^2(\alpha^{2i} E[e_{i-1}^2] + \sigma_w^2)}{\sigma_w^2\, E[e_{i-1}^2]} = \alpha^{2i} + K_{i-1}.$$

Substituting back into (1):

$$\hat{\theta}_i = (1 - l_i\alpha^i)\hat{\theta}_{i-1} + l_i\, y_i = \frac{\sigma_w^2}{\alpha^{2i} E[e_{i-1}^2] + \sigma_w^2}\,\hat{\theta}_{i-1} + \frac{E[e_{i-1}^2]\,\alpha^i}{\alpha^{2i} E[e_{i-1}^2] + \sigma_w^2}\, y_i.$$

Multiplying numerator and denominator by K_i:

$$\boxed{\hat{\theta}_i = K_i^{-1}\Big[K_{i-1}\,\hat{\theta}_{i-1} + \alpha^i\, y_i\Big].}$$

(c) Behavior of e_i

From part (b), $K_i = K_{i-1} + \alpha^{2i}$, so unrolling:

$$K_i = K_0 + \sum_{k=1}^{i} \alpha^{2k} = \frac{\sigma_w^2}{\sigma^2} + \sum_{k=1}^{i} \alpha^{2k}.$$

Therefore:

$$\frac{\sigma_w^2}{e_i} = \frac{\sigma_w^2}{\sigma^2} + \sum_{k=1}^{i} \alpha^{2k} \qquad \Rightarrow \qquad e_i = \frac{1}{\dfrac{1}{\sigma^2} + \dfrac{1}{\sigma_w^2}\displaystyle\sum_{k=0}^{i} \alpha^{2k}}.$$

- When $|\alpha| \geq 1$: $\sum_{k=0}^{i} \alpha^{2k} \to \infty$, so $e_i \to 0$ as $i \to \infty$.
- When $|\alpha| < 1$: $\sum_{k=0}^{\infty} \alpha^{2k} = \dfrac{1}{1-\alpha^2}$, so

$$e_i \xrightarrow{i\to\infty} \frac{1}{\dfrac{1}{\sigma^2} + \dfrac{\alpha^2}{\sigma_w^2(1-\alpha^2)}}.$$

Problem 2

Ans:

(a) MMSE estimate of x_2 given x_1

Given:

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{x_1}^2 & \mathrm{cov}(x_1, x_2) \\ \mathrm{cov}(x_1, x_2) & \sigma_{x_2}^2 \end{bmatrix} \right).$$

From the conditional Gaussian result:

$$x_2 \mid x_1 \sim \mathcal{N}\left(\frac{\mathrm{cov}(x_1, x_2)}{\sigma_{x_1}^2} x_1,\ \sigma_{x_2}^2 - \frac{[\mathrm{cov}(x_1, x_2)]^2}{\sigma_{x_1}^2} \right).$$

Therefore:

$$\hat{x}_2(x_1) = E[x_2 \mid x_1] = \frac{\mathrm{cov}(x_1, x_2)}{\sigma_{x_1}^2} x_1.$$

Since the MMSE estimate is linear in x_1, MMSE = LLS in this case.

(b) Cost and singular R_x

$$C(x_2, \hat{x}_2) = E[(x_2 - \hat{x}_2)^2] = E_{x_1}[\mathrm{Var}(x_2 \mid x_1)] = \sigma_{x_2}^2 - \frac{[\mathrm{cov}(x_1, x_2)]^2}{\sigma_{x_1}^2}.$$

Now $C(x_2, \hat{x}_2) = 0 \iff \sigma_{x_1}^2 \sigma_{x_2}^2 = [\mathrm{cov}(x_1, x_2)]^2$, which is exactly the condition for R_x to be singular.

(c) Perfect estimation when R_x is singular

Suppose we want to estimate x_1 given $x_2, \ldots, x_N$ (WLOG). Let $\mathbf{x}' = [x_2, \ldots, x_N]^\top$. The LLS estimate is:

$$\hat{x}_1 = \Sigma_{x_1 \mathbf{x}'} \Sigma_{\mathbf{x}'\mathbf{x}'}^{-1} \mathbf{x}',$$

with cost:

$$C(x_1, \hat{x}_1) = \sigma_{x_1 x_1} - \Sigma_{x_1 \mathbf{x}'} \Sigma_{\mathbf{x}'\mathbf{x}'}^{-1} \Sigma_{\mathbf{x}' x_1}.$$

Since $R_x \succeq 0$ and $\det(R_x) = 0$, using the Schur complement formula:

$$\det(R_x) = \det(\Sigma_{\mathbf{x}'\mathbf{x}'}) \cdot \left(\sigma_{x_1 x_1} - \Sigma_{x_1 \mathbf{x}'} \Sigma_{\mathbf{x}'\mathbf{x}'}^{-1} \Sigma_{\mathbf{x}' x_1} \right) = 0.$$

This implies $\sigma_{x_1 x_1} - \Sigma_{x_1 \mathbf{x}'} \Sigma_{\mathbf{x}'\mathbf{x}'}^{-1} \Sigma_{\mathbf{x}' x_1} = 0$, i.e. $C(x_1, \hat{x}_1) = 0$.

Hence **perfect estimation is possible**.

Problem 3

Ans:

Setup: $y = x + w$, $x \sim \text{Uniform}(0, \alpha)$, $w \sim \text{Exp}(\lambda)$, independent.

(a)(i) Conditional pdf $f(y \mid x)$

$$P(y \le t \mid x) = P(w \le t - x \mid x) = F_w(t - x).$$
$$f(y \mid x) = f_w(y - x) = \lambda e^{-\lambda(y-x)}\, \mathbf{1}_{\{y \ge x\}}.$$

Support: $y \ge x$.

(a)(ii) Marginal pdf $f(y)$

$$f(y) = \int_{-\infty}^{\infty} f(y \mid x)\, f(x)\, dx = \frac{\lambda}{\alpha} \int_0^{\alpha} e^{-\lambda(y-x)}\, \mathbf{1}_{\{y \ge x\}}\, dx.$$

Case 1: $0 \le y \le \alpha$

$$f(y) = \frac{\lambda}{\alpha} \int_0^{y} e^{-\lambda(y-x)}\, dx = \frac{\lambda}{\alpha}\, e^{-\lambda y}\, \frac{e^{\lambda y} - 1}{\lambda} = \frac{1}{\alpha}(1 - e^{-\lambda y}).$$

Case 2: $y \ge \alpha$

$$f(y) = \frac{\lambda}{\alpha} \int_0^{\alpha} e^{-\lambda(y-x)}\, dx = \frac{\lambda}{\alpha}\, e^{-\lambda y}\, \frac{e^{\lambda \alpha} - 1}{\lambda} = e^{-\lambda y}\, \frac{e^{\lambda \alpha} - 1}{\alpha}.$$

$$f(y) = \begin{cases} 0 & y < 0, \\ \dfrac{1}{\alpha}(1 - e^{-\lambda y}) & 0 \le y \le \alpha, \\ e^{-\lambda y}\, \dfrac{e^{\lambda \alpha} - 1}{\alpha} & y \ge \alpha. \end{cases}$$

(a)(iii) Posterior pdf $f(x \mid y)$

$$f(x \mid y) = \frac{f(y \mid x)\, f(x)}{f(y)} = \frac{\lambda\, e^{\lambda x}\, \mathbf{1}_{\{0 \le x \le \min(y,\alpha)\}}}{(e^{\lambda y} - 1)\, \mathbf{1}_{\{0 \le y \le \alpha\}} + (e^{\lambda \alpha} - 1)\, \mathbf{1}_{\{y \ge \alpha\}}}.$$

Support: $0 \le x \le \min(y, \alpha)$.

(b) MAP estimate

$$\hat{x}_{\text{MAP}} = \arg\max_x\, f(x \mid y).$$

Since $\lambda e^{\lambda x}$ is increasing in x for $\lambda > 0$, the maximum is at the right boundary of the support:

$$\boxed{\hat{x}_{\text{MAP}} = \min(y, \alpha).}$$

(c) MAE (Median) estimate

$\hat{x}_{\text{MAE}}$ is the median of $f(x \mid y)$, satisfying $\int_{\hat{x}_{\text{MAE}}}^{\min(y,\alpha)} f(x \mid y)\, dx = \frac{1}{2}$.

Case 1: $\min(y, \alpha) = y$ *(i.e.* $y \leq \alpha$*)*

$$\int_{\hat{x}_{\text{MAE}}}^{y} \frac{\lambda e^{\lambda x}}{e^{\lambda y} - 1}\, dx = \frac{1}{2} \implies \frac{e^{\lambda y} - e^{\lambda \hat{x}_{\text{MAE}}}}{e^{\lambda y} - 1} = \frac{1}{2} \implies \hat{x}_{\text{MAE}} = \frac{1}{\lambda} \ln \frac{e^{\lambda y} + 1}{2}.$$

Case 2: $\min(y, \alpha) = \alpha$ *(i.e.* $y \geq \alpha$*)*

$$\int_{\hat{x}_{\text{MAE}}}^{\alpha} \frac{\lambda e^{\lambda x}}{e^{\lambda \alpha} - 1}\, dx = \frac{1}{2} \implies 2(e^{\lambda \alpha} - e^{\lambda \hat{x}_{\text{MAE}}}) = e^{\lambda \alpha} - 1 \implies \hat{x}_{\text{MAE}} = \frac{1}{\lambda} \ln \frac{e^{\lambda \alpha} + 1}{2}.$$

Combining:

$$\boxed{\hat{x}_{\text{MAE}} = \frac{1}{\lambda} \ln \frac{e^{\lambda \min(y,\alpha)} + 1}{2}.}$$

(d) MMSE estimate

$$\hat{x}_{\text{MMSE}} = E[x \mid y] = \int_{0}^{\min(y,\alpha)} x \cdot \frac{\lambda e^{\lambda x}}{e^{\lambda \min(y,\alpha)} - 1}\, dx.$$

Let $b = \lambda \min(y, \alpha)$ and substitute $t = \lambda x$:

$$= \frac{1}{\lambda^2 (e^b - 1)} \int_0^b t\, e^t\, dt = \frac{1}{\lambda^2 (e^b - 1)} \left[t e^t - e^t \right]_0^b = \frac{1}{\lambda^2 (e^b - 1)} \left[e^b (b - 1) + 1 \right].$$

$$\boxed{\hat{x}_{\text{MMSE}} = \frac{e^{\lambda \min(y,\alpha)} \big(\lambda \min(y, \alpha) - 1 \big) + 1}{\lambda \big(e^{\lambda \min(y,\alpha)} - 1 \big)}.}$$

Problem 4

Ans:

The system model is the Gauss–Markov process:

$$x_{i+1} = A\, x_i + B\, w_i, \qquad y_i = C\, x_i + v_i,$$

where $w_i \sim \mathcal{N}(0, W)$ and $v_i \sim \mathcal{N}(0, V)$ are independent process and measurement noise, and $x_0 \sim \mathcal{N}(\hat{x}_0, \Sigma_0)$.

(a) Helper function: sampling from a multivariate Gaussian

To simulate the state and measurement trajectories, we need to draw samples from multivariate normal distributions at each time step. The following one-line function wraps NumPy's sampler and returns a column vector:

```python
import numpy as np

def sample_from_multivariate_normal(mean, cov):
    return np.random.multivariate_normal(mean, cov, 1).T
```

Explanation. `np.random.multivariate_normal(mean, cov, 1)` draws one sample from $\mathcal{N}(\texttt{mean}, \texttt{cov})$ and returns a row vector of shape $(1, n)$. Transposing with `.T` converts it to a column vector of shape $(n, 1)$, which matches the standard state-space convention $x_i \in \mathbb{R}^n$.

(b) Kalman Filter Trajectory: True State vs. Filtered Estimate

The Kalman filter was run for $K = 100$ time steps with measurement noise variance $V \in \{0.1, 0.2, \ldots, 0.9\}$ (the case $V = 1.0$ is omitted here as it is covered in part (c)). Figure 1 shows the first component $x_{k,1}$ of the true state alongside the filtered estimate $\hat{x}_{k|k-1,1}$ for each value of V.

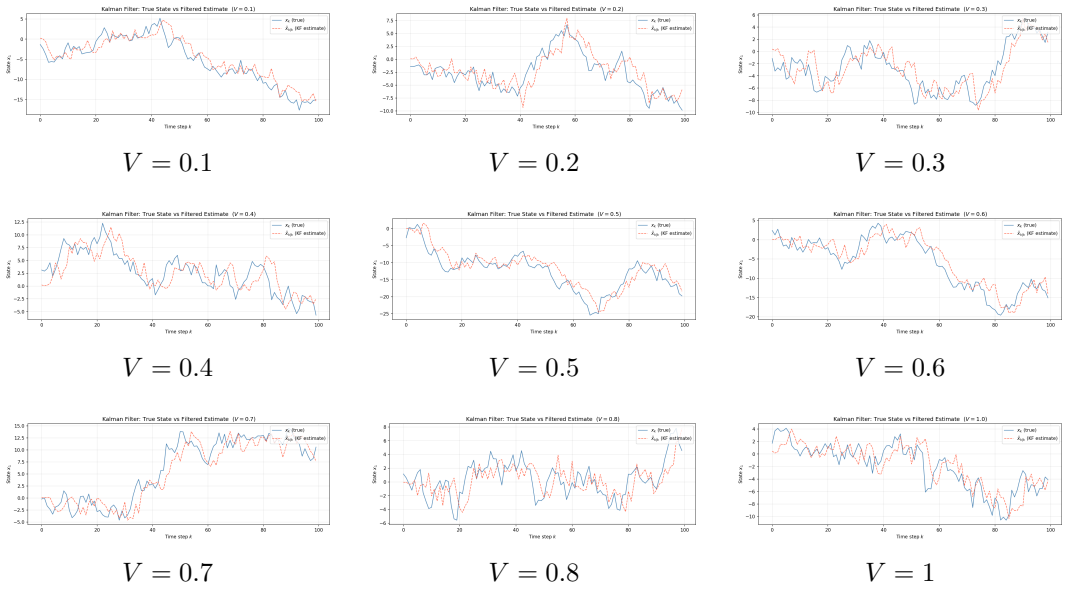

Figure 1: True state $x_{k,1}$ (blue) vs. KF predicted estimate $\hat{x}_{k|k-1,1}$ (red dashed) for varying measurement noise variance V.

Discussion. As V decreases, the measurement noise decreases and the measurements become more informative. Consequently, the filter trusts the observations more and the predicted estimate tracks the true state more closely. At small V (e.g. $V = 0.1$), the two curves are nearly indistinguishable, while at large V (e.g. $V = 1$) the estimate lags noticeably behind the true state.

(c) Theoretical vs. Empirical Error Covariance

The theoretical prediction error covariance $R_{k|k-1}$ is propagated via the Riccati recursion:

$$R_{k|k-1} = A\, R_{k-1|k-1}\, A^\top + B\, W\, B^\top, \qquad R_{k|k} = R_{k|k-1} - L_k\, C\, R_{k|k-1},$$

and compared against the empirical trace obtained from $M = 100{,}000$ independent Monte Carlo runs at $V = 1$.

The predicted error covariance is defined as:

$$R_{k|k-1} = E\big[(x_k - \hat{x}_{k|k-1})(x_k - \hat{x}_{k|k-1})^\top\big]\,,$$

so its trace equals the total mean squared prediction error:

$$\mathrm{tr}(R_{k|k-1}) = E\big[\|x_k - \hat{x}_{k|k-1}\|^2\big]\,.$$

To estimate this empirically, the KF is run M times independently. On the i-th run the prediction error at time k is $e_k^{(i)} = x_k^{(i)} - \hat{x}_{k|k-1}^{(i)}$, and the sample mean

$$\widehat{\mathrm{tr}(R_{k|k-1})} = \frac{1}{M} \sum_{i=1}^{M} \left(e_k^{(i)}\right)^\top e_k^{(i)}$$

converges to $\mathrm{tr}(R_{k|k-1})$ as $M \to \infty$ by the Law of Large Numbers.

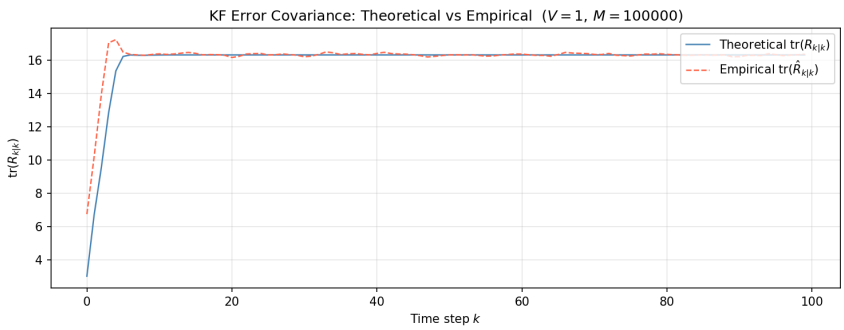

Figure 2: Theoretical $\mathrm{tr}(R_{k|k-1})$ (blue) vs. empirical $\mathrm{tr}(\hat{R}_{k|k-1})$ (red dashed) for $V = 1$, $M = 100{,}000$.

Discussion. The Riccati recursion converges to the steady-state solution of the algebraic Riccati equation (ARE), so $\mathrm{tr}(R_{k|k-1})$ saturates after a few dozen time steps. The empirical trace closely follows the theoretical curve, with the match improving as k increases because the initial transient (sensitive to the choice of Σ_0) dies out and the filter reaches its steady state.

Problem 5

Ans:

(a) Linearity of LLS

From the LLS derivation, with $\beta = A_1\theta_1 + A_2\theta_2 + c$:

$$\hat{\beta}_{\mathrm{LLS}}(\mathbf{y}) = \mu_\beta + R_{\beta y}\, R_{yy}^{-1}(\mathbf{y} - \mu_y).$$

$$\mu_\beta = A_1\mu_{\theta_1} + A_2\mu_{\theta_2} + c.$$

$$
\begin{aligned}
R_{\beta y} &= E[(\beta - \mu_\beta)(\mathbf{y} - \mu_y)^\top] \\
&= E\big[(A_1(\theta_1 - \mu_{\theta_1}) + A_2(\theta_2 - \mu_{\theta_2}))(\mathbf{y} - \mu_y)^\top\big] \\
&= A_1 R_{\theta_1 y} + A_2 R_{\theta_2 y}.
\end{aligned}
$$

Therefore:

$$
\begin{aligned}
\hat{\beta}_{\mathrm{LLS}}(\mathbf{y}) &= A_1\mu_{\theta_1} + A_1 R_{\theta_1 y} R_{yy}^{-1}(\mathbf{y} - \mu_y) + A_2\mu_{\theta_2} + A_2 R_{\theta_2 y} R_{yy}^{-1}(\mathbf{y} - \mu_y) + c \\
&= A_1\hat{\theta}_{1,\mathrm{LLS}} + A_2\hat{\theta}_{2,\mathrm{LLS}} + c. \quad \square
\end{aligned}
$$

(b) Orthogonal observations

If y_1 and y_2 are orthogonal, $E[y_1 y_2^\top] = 0$, so R_{yy} is block-diagonal:

$$
R_{yy} = \begin{bmatrix} \Sigma(y_1, y_1) & 0 \\ 0 & \Sigma(y_2, y_2) \end{bmatrix}, \qquad R_{yy}^{-1} = \begin{bmatrix} \Sigma^{-1}(y_1, y_1) & 0 \\ 0 & \Sigma^{-1}(y_2, y_2) \end{bmatrix}.
$$

Therefore:

$$
\begin{aligned}
\hat{x}_{\mathrm{LLS}}(y_1, y_2) &= R_{xy}\, R_{yy}^{-1} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \\
&= \begin{bmatrix} \Sigma(x, y_1) & \Sigma(x, y_2) \end{bmatrix} \begin{bmatrix} \Sigma^{-1}(y_1, y_1) & 0 \\ 0 & \Sigma^{-1}(y_2, y_2) \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \\
&= \Sigma(x, y_1)\, \Sigma^{-1}(y_1, y_1)\, y_1 + \Sigma(x, y_2)\, \Sigma^{-1}(y_2, y_2)\, y_2 \\
&= \hat{x}_{\mathrm{LLS}}(y_1) + \hat{x}_{\mathrm{LLS}}(y_2). \quad \square
\end{aligned}
$$